BSH/S-24/CS2T007/03

JAIDEV EDUCATION SOCIETY'S

# J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

## B. Tech Summer / ~~Winter~~ - 2024

## (ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

**Branch: Computer Science & Engineering** **Sem:- II**

**Subject: VSEC (Networking Tools & Techniques)** **Subject Code: CS2T007**

**Marks: 50** **Date of Exam: 20/06/2024** **Time: 2 Hrs.**

**Instructions to the students:**

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

**COs: Students should be able to:**

1. Identify the operation and functionality of computer networks and how they are used in modern business environments.
2. Demonstrate proficiency in configuring, managing, and troubleshooting switches.
3. Apply troubleshoot common network problems, including connectivity issues, routing problems and security breaches.
4. Get an extensive knowledge of network management and communication switching techniques.
5. Implement practical and real time use of different network operating tools and identifying basic security threats.

## SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.1 | Define Computer Network. | L1/CO1 | 2 |
| Q.2 | List the types of Network Applications. | L1/CO1 | 2 |
| Q.3 | Enlist the different Networking Devices. | L1/CO2 | 2 |
| Q.4 | State the advantages of Local Area Networks. | L1/CO2 | 2 |
| Q.5 | Write short note on Ethernet. | L3/CO5 | 2 |

## SECTION B

1. **All Questions are compulsory.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.6 | Distinguish between Circuits Switching Vs Packet Switching. | L4/CO4 | 4 |
| Q.7 | Explain the modes of VLAN Trunking Protocols. | L2/CO3 | 4 |
| Q.8 | Write short notes on IPERF and JPERF. | L5/CO4 | 4 |
| Q.9 | Draw & Discuss Network Structure and Architecture. | L4/CO5 | 4 |

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|---|---|---|---|
| Q.10 | Illustrate Network Access Control in detail. | L3/CO4 | 4 |
| Q.11 | Examine TCP/IP Model with its layers. | L4/CO1 | 4 |
| Q.12 | Analyse OSI reference Model with diagram. | L4/CO1 | 4 |

## SECTION C

1. **Solve any TWO Questions.**

| Q. No. | Questions | Level/CO Mapped | Marks |
|---|---|---|---|
| Q.13 | Illustrate ISP with its advantages and disadvantages. | L4/CO4 | 6 |
| Q.14 | Explain topology with different types of physical Network Topologies. | L5/CO3 | 6 |
| Q.15 | Describe Network Protocols with its types in detail. | L6/CO5 | 6 |



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